Econometrics Chapter 1: A Recap on Required Mathematical Statistics

1.1 Mathematics Review

Statistic

- field of study: estimate or learn certain properties about a population given a sample

Probability Theory

- studies how likely events are to occur under uncertainty

(1) Summation Prerequisites

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \dots + x_n$$

whereby, $\{x_i\}_{i=1}^{n} = \{x_1, x_2, \dots, x_n\}$

Summation Properties

1. $\sum_{i=1}^{n} c = nc$
2. $\sum_{i=1}^{n} cx_i = c\sum_{i=1}^{n} x_i$, where c is a constant value
3. $\sum_{i=1}^{n} (x_i + y_i) = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i$
4. $\sum_{i=1}^{n} (x_i - y_i) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} y_i$
5. $\sum_{i=1}^{n} x_i = \sum_{i=1}^{k} x_i + \sum_{i=k+1}^{n} x_i$ for $1 \le k \le n$

★ Two Important Summation Results ★

1. Sum of deviations from Mean is Zero

$$\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x}$$

$$= n\bar{x} - n\bar{x}$$

$$= 0$$

Note: $\sum_{i=1}^{n} x_i = n\bar{x}$ or $\frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$

2. Alternate Expressions: Sum of Products of deviation from Means

$$\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}) = \sum_{i=1}^{n}(x_i-\bar{x})y_i - (x_i-\bar{x})\bar{y}$$

$$= \sum_{i=1}^{n}(x_i-\bar{x})y_i - \bar{y}\underbrace{\sum_{i=1}^{n}(x_i-\bar{x})}_{0}$$

$$= \sum_{i=1}^{n}(x_i-\bar{x})y_i$$

note: this is equal to $\sum_{i=1}^{n}(y_i-\bar{y})x_i$

$$= \sum_{i=1}^{n}x_iy_i - \bar{x}\sum_{i=1}^{n}y_i$$

$$= \sum_{i=1}^{n}x_iy_i - n\bar{x}\bar{y}$$

(2) Probabilities Pre-requisites

Random Variables ——— Discrete or Continuous

- numerical quantity whose value depends on the outcome of a random process
- can be independent or dependent

Expectation Properties:

1. $E(X) = \mu = \sum x\, p(x)$
2. $E(c) = c$ (c constant)
3. $E(cX) = cE(X)$
4. $E(E(X)) = E(X) = \mu$
5. $E(X+Y) = E(X) + E(Y)$
6. $E(aX+b) = aE(X) + b$
7. $E(XY) = E(X)E(Y)$ Independent
8. $E(XY) = E(X)E(Y) + Cov(X, Y)$ Dependent

Variance Properties

1. $Var(X) = E(X-E(X))^2 = E(X^2) - (E(X))^2$
2. $Var(c) = 0$
3. $Var(X+c) = Var(X)$
4. $Var(X-c) = Var(X)$
5. $Var(cX) = c^2 Var(X)$
6. $Var\left(\frac{X}{c}\right) = \frac{1}{c^2} Var(X)$
7. $Var(aX+b) = a^2 Var(X)$
8. Independent Random Variable

$$Var(X_1 + X_2 + \dots + X_n) = Var(X_1) + Var(X_2) + \dots + Var(X_n)$$

Dependent Random Variable

$$Var(aX+bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\, Cov(X, Y)$$

Covariance Properties

1. $Cov(X, Y) = E(X - E(X))(Y - E(Y))$

   $= E(XY) - E(X)E(Y)$

2. $Cov(X, X) = Var(X)$

3. $Cov(aX + b, cY + d) = ac\,Cov(X, Y)$

4. $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$

(3) Types of Distributions

Discrete Random Variables [Eg. no. of students]

- takes countable values
- probabilities assigned to points

Continuous Random Variables [Eg. height of students]

- within an interval
- probabilities assigned to areas under a curve

Probability Distribution Functions (PDF)

- describes how probabilities of a random variable are assigned across its possible values

(i) Probability Mass Function —— for discrete random variables

—— $f_X(x) = P(X = x)$

(ii) Probability Density Function —— for continuous random variables

—— $P(a \leq X \leq b) = \int_a^b f(x)\,dx$

Types of Distributions for discrete RV [Each describes a different kind of random process]

1. Bernoulli

Describes the random process behind Y

- two possible outcomes: $x = 0, 1$

$P(X=1) = p \quad P(X=0) = 1-p$

$P(X=x)$ where $x = 0, 1$

$f_x(x) = p^x (1-p)^{1-x}$ — purpose: $\prod_{i=1}^{n} p^{x_i} (1-p)^{1-x_i}$

2. Binomial — same bernoulli experiment n times, independently with the same p.

- number of successes in n independent Bernoulli trials
- $P(X=x) = \binom{n}{x} p^x q^{n-x}$ whereby $q = 1-p$

whereby,

p — probability of success in one trial

q — probability of failure in one trial

n — no. of trial

x — no. of success desired

- Mean : $\mu = E(x) = p$

Variance : $\sigma^2 = np(1-p)$

3. Poisson

- models the number of times an event occurs in a fixed interval of time

when • events occur independently

• constant average rate λ

• cannot occur at the same time

- $P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$, $k = 0, 1, 2 \ldots$

predicted mean count: $E[Y|X=x]$

$Y|X=x \sim poisson(\lambda(x))$, set $\lambda(x) = \exp(x'\beta)$ $= \exp(x'\beta)$

Y — how many times something happens in a fixed period/space

• Eg. no. of emails an hour

- poisson assumes mean = variance
- Binomial $(n,p) \approx$ Poisson $(\lambda = np)$ when n is big, p is small

Types of Distributions for Continuous RV

1. Gaussian Distribution

- mean : $\mu = E(x) = \sum x_i P(x)$

Bell Curve : $f_x(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), x \in \mathbb{R}$

Variance : $\sigma^2 = E(X - E(X))^2 = E(X^2) - (E(X))^2$

- known variance σ^2

$Z = \frac{X-\mu}{\sigma}$

Normal (μ, σ^2)

2. Student t-distribution

$t = \frac{\bar{X}-\mu_0}{s/\sqrt{n}}$

- unknown σ^2
- bell shape, fatter tails than gaussian distribution (Higher Probability of extreme value)

3. Chi-Square χ^2

- used to test spread of variances

Example (econometrics-friendly): income volatility after a policy change

Say you're studying a job-training program.

- Group A: people **before** training
- Group B: people **after** training

You might find the *mean* monthly income is similar, but you also care whether the program makes income **more stable**.

So you compare:

$$Var(\text{income}_A) \quad \text{vs} \quad Var(\text{income}_B)$$

- If $Var(\text{income}_B)$ is **smaller**, incomes are less volatile → the program improved stability.
- If it's **larger**, maybe the program pushes people into commission-based jobs → more risk.

4. F Distribution

- ratio of two scaled Chi-square variables
- Test if two populations have the same variances

5. ANOVA

- statistical method used to compare the means of three or more groups to see if at least one mean is significantly different from others

(4) Statistical Review

Random Sampling

- every individual has an equal chance of getting selected
- an individual sampled does not make another more or less likely to be sampled

Statistical Model [Population Model]

- allow $\{Y_i\}_{i=1}^n$ to be the randomly sampled sample

$Y_i \overset{iid}{\sim} \text{Normal}(\mu, \sigma^2)$

whereby,

iid – independently and identically distributed

- $E(Y_i) = \mu$

$Var(Y_i) = \sigma^2$

Criterion for Good Estimators:

1. Unbiasedness

$E(\hat{\theta}) = \theta$

2. Efficiency

- covered in (5)

Statistical Estimator of Population

(i) Estimators of Population Mean

1. Estimator 1: $\hat{\mu} = \bar{Y} = \frac{1}{n}\sum Y_i$

Bias Test: $E(\bar{Y}) = E\left(\frac{1}{n}\sum Y_i\right)$

$= \frac{1}{n} E\left(\sum Y_i\right)$

$= \frac{1}{n} E(n\mu)$

$= \mu$

$\therefore \bar{Y}$ is an unbiased estimator of μ

2. Estimator 2: $\tilde{\mu} = \sum \omega_i Y_i$ whereby, ω_i is the weightage of each value Y_i

Bias test: $E(\tilde{\mu}) = E(\sum \omega_i Y_i)$

In this case $\omega_i = \frac{1}{n}$ → does not change randomly since ω_i are constants and not random variables

$= \sum \omega_i E(Y_i)$

$= \sum \omega_i \mu$

$= \mu \sum \omega_i$

$\therefore$ For $\tilde{\mu}$ to be an unbiased estimator of μ, $\sum \omega_i = 1$

(ii) Estimators of Population Variance

1. Estimator 1: $\tilde{\sigma}^2 = \frac{1}{n}\sum (Y_i - \bar{Y})^2$ ★Bias Estimate

$= \frac{1}{n}\sum (Y_i - \bar{Y})(Y_i - \bar{Y})$

$= \frac{1}{n}\left[\sum Y_i Y_i - n\bar{Y}\bar{Y}\right]$

$= \frac{1}{n}\sum Y_i^2 - \bar{Y}^2$

Bias Test: $E(\tilde{\sigma}^2) = E\left[\frac{1}{n}\sum Y_i^2 - \bar{Y}^2\right]$

$= \frac{1}{n} E(\sum Y_i^2) - E(\bar{Y}^2)$

Part 1: $\frac{1}{n} E(\sum Y_i^2)$

$Var(Y_i) = E(Y_i^2) - (E(Y_i))^2$

$\sigma^2 = E(Y_i^2) - \mu^2$

$E(Y_i^2) = \sigma^2 + \mu^2$

Part 2: $E(\bar{Y}^2)$

$$Var(\bar{Y}) = E(\bar{Y}^2) - [E(\bar{Y})]^2$$

$$Var\left(\frac{1}{n}\sum Y_i\right) = E(\bar{Y}^2) - \mu^2$$

Part 3: $Var\left(\frac{1}{n}\sum Y_i\right) = \frac{1}{n^2} Var\left(\sum Y_i\right)$

$$= \frac{1}{n^2}(n\sigma^2)$$

$$= \frac{1}{n}\sigma^2$$

Putting All Together

$$E(\tilde{\sigma}^2) = \sigma^2 + \mu^2 - \left(\frac{1}{n}\sigma^2 + \mu^2\right)$$

$$= \left(1 - \frac{1}{n}\right)\sigma^2$$

$$= \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

$\therefore \tilde{\sigma}^2$ is a biased estimator of σ^2

2. Estimator 2: $\hat{\sigma}^2 = \frac{n}{n-1}\tilde{\sigma}^2$

$$= \frac{1}{n-1}\sum(Y_i - \bar{Y})^2$$

Bias Test: $E(\hat{\sigma}^2) = \frac{n}{n-1} E(\tilde{\sigma}^2)$

$$= \frac{n}{n-1} \times \frac{n-1}{n}\sigma^2$$

$$= \sigma^2$$

$\therefore \hat{\sigma}^2$ is an unbiased estimator of σ^2

n-1 degree of freedom

Estimator of Standard Error of Statistical Estimators

(i) Standard Error of Sample Mean

- Many Sample Sets, with many sample means
- SE of Sample Mean is the Standard deviation of these means

$$\begin{aligned}\text{Var}(\bar{Y}) &= \text{Var}\left(\frac{1}{n}\sum Y_i\right) \\ &= \frac{1}{n^2}\text{Var}\left(\sum Y_i\right) \\ &= \frac{1}{n^2}(n\sigma^2) \\ &= \frac{\sigma^2}{n}\end{aligned}$$

$\widehat{\text{Var}}(\bar{Y}) = \frac{\hat{\sigma}^2}{n}$ since most of the time σ^2 is unknown

$SE(\bar{Y}) = \sqrt{\frac{\hat{\sigma}^2}{n}}$

(ii) Standard Error of Sample Variance

- 3 Methods:

Method 1: Assume Normal Distribution $Y_i \overset{iid}{\sim} (\mu, \sigma^2)$

- if Y_i follows a normal distribution

The distribution of SE follows a chi-square distribution

$\frac{(n-1)\hat{\sigma}^2}{\sigma^2} \sim \chi_{(n-1)}$ which has a variance of $2(n-1)$

$$\text{Var}\left(\frac{(n-1)\hat{\sigma}^2}{\sigma^2}\right) = \frac{(n-1)^2}{\sigma^4}\text{Var}(\hat{\sigma}^2)$$

$$\frac{(n-1)^2}{\sigma^4}\text{Var}(\hat{\sigma}^2) = 2(n-1)$$

$$\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n-1}$$

$$S.E(\hat{\sigma}^2) = \sqrt{\frac{2\sigma^4}{n-1}}$$

Method 2: Without assuming normality

$\text{Var}(\hat{\sigma}^2) = \frac{1}{n}\mu_4 - \frac{n-3}{n-1}\sigma^4$ whereby $\mu_4 = E(Y - E(Y))^4$

Proof: $\text{Var}(\hat{\sigma}^2) = \text{Var}\left(\frac{1}{n-1}\Sigma(Y_i - \bar{Y})^2\right)$

★ HW ★

Method 3: Bootstrap

- idea of bootstrapping:

Resample from $\{Y_1, \ldots, Y_n\}$ with replacement

to obtain bootstrap sample $\{Y_1^{(b)}, \ldots, Y_n^{(b)}\}$ for $b = 1, \ldots, B$

- Calculate $\hat{\delta}_2^{(b)}$ for each bootstrap sample

$$\text{Var}(\hat{\delta}_2) = \frac{1}{B-1} \sum_{i=1}^{B} (\hat{\delta}_2^{(b)} - \bar{\delta}_2^{(b)})$$

$$\text{S.E}(\hat{\delta}_2) = \sqrt{\frac{1}{B-1} \sum_{i=1}^{B} (\hat{\delta}_2^{(b)} - \bar{\delta}_2^{(b)})}$$

(5) Concept of Efficiency

Efficiency

- smaller estimator variance is preferred

Example: Comparing $\tilde{\mu}$ (weighted mean) and $\hat{\mu}$ (average mean)

Weighted Mean:

$$\text{Var}(\tilde{\mu}) = \text{Var}(\Sigma \omega_i Y_i) \quad \text{allow} \quad \omega_i = \frac{1}{n} + v_i$$

$$= \text{Var}(\Sigma(\tfrac{1}{n} + v_i) Y_i)$$

$$= \text{Var}(Y_i) \Sigma(\tfrac{1}{n} + v_i)^2$$

$$= \sigma^2 \Sigma(\tfrac{1}{n^2} + \tfrac{2v_i}{n} + v_i^2)$$

$$= \sigma^2(\tfrac{1}{n^2} + \tfrac{2v_i}{n} + v_i^2)$$

$$= \tfrac{\sigma^2}{n^2} + \tfrac{2v_i}{n} + v_i^2 \geq \text{Var}(\hat{\mu})$$

Average Mean:

$$\text{Var}(\hat{\mu}) = \text{Var}(\tfrac{1}{n} \Sigma Y_i)$$

$$= \tfrac{1}{n^2}(n\sigma^2)$$

$$= \tfrac{\sigma^2}{n} < \text{Var}(\tilde{\mu})$$

Mean Square Estimator Error — Another method of efficiency

- quality of estimators can be tested by Mean Square Estimator Error (MSE)
- $MSE(\hat{\theta}) = E((\hat{\theta} - \theta)^2)$

$= Var(\hat{\theta} - \theta) + (E(\hat{\theta} - \theta))^2$

$=$ Estimator variance + estimator bias

- choose estimator to minimize MSE allowing for bias-variance trade off

Exercise: Show that if $Y_i \overset{iid}{\sim}$ Normally (μ, σ^2)

then $MSE(\tilde{\sigma}^2) < MSE(\hat{\sigma}^2)$

Solution

- Assume both distribution of estimators follows a chi square distribution

Then,

$Var(\hat{\sigma}^2) = Var(\tilde{\sigma}^2) = \frac{2\sigma^4}{n-1}$

Bias Test

$E(\tilde{\sigma}^2) = \frac{n-1}{n}\sigma^2$

$E(\hat{\sigma}^2) = \sigma^2$

$MSE(\tilde{\sigma}^2) = \frac{n-1}{n}\sigma^2 + \frac{2\sigma^4}{n-1}$

$MSE(\hat{\sigma}^2) = \sigma^2 + \frac{2\sigma^4}{n-1}$

$MSE(\tilde{\sigma}^2) < MSE(\hat{\sigma}^2)$ since $\frac{n-1}{n}\sigma^2 < \sigma^2$ (Shown)

(6) Consistency

Consistency

- an estimator is consistent if it converges in probability to what it is estimating
- mathematical representation:

Given a sequence of Random Variables $X_n, n=1,2,\dots,\infty$

$P(|X_n - c| \geq \varepsilon) = 0$

converges in probability to c if for any $\varepsilon > 0$

$X_n \xrightarrow{p} c$

Properties of Convergent in Probabilities

1. $X_n \xrightarrow{p} Y_n$

   Then, $X_n - Y_n \xrightarrow{p} 0$

2. If $g(\cdot)$ is continuous and $X_n \xrightarrow{p} c$

   then $g(X_n) \to g(c)$

Khinchine's Weak Law of Large Numbers (WLLN)

- states that Sample mean $\hat{\mu}$ is consistent for population mean

If $\{X_i\}_{i=1}^n$ is iid with $E(X_i) = \mu < \infty$ for all i, then

$\bar{X}_n \xrightarrow{p} \mu$

Application : Estimation of μ^2

- Since $\bar{Y} \xrightarrow{p} \mu$

  then $\bar{Y}^2 \xrightarrow{p} \mu^2$

- Note: While $\bar{Y}^2$ converges in probability to μ^2

  It is not an unbiased estimator of μ^2

$$\text{Var}(\bar{Y}) = E(\bar{Y}^2) - (E(\bar{Y}))^2$$

$$\frac{\sigma^2}{n} = E(\bar{Y}^2) - \mu^2$$

$$E(\bar{Y}^2) = \frac{\sigma^2}{n} + \mu^2 \neq \mu^2$$

Jensen's Inequality

- If $g(\cdot)$ is convex then $E(g(X)) \geq g(E(X))$
- If $g(\cdot)$ is concave then $E(g(X)) \leq g(E(X))$
- Equality holds if $g(\cdot)$ is linear

Example 1: Show $\tilde{\sigma}^2$ is consistent for σ^2

Sol^n: $\tilde{\sigma}^2 = \frac{1}{n}\sum(Y_i - \bar{Y})^2$

$= \frac{1}{n}[E(Y_i^2) - E(\bar{Y})^2]$

$= \frac{1}{n}[\sigma^2 + \mu^2 - \mu^2)$

$= \frac{1}{n}\sigma^2$

As $n \to \infty$, $\frac{1}{n} \to 1$

$\tilde{\sigma}^2 = \frac{1}{n}\sigma^2 \xrightarrow{P} \sigma^2$ (Shown)

Example 2: Show that $\hat{\sigma}^2$ is consistent for σ^2

$\hat{\sigma}^2 = \frac{n-1}{n}(\tilde{\sigma}^2)$

$\frac{n-1}{n} \to 1$, $\tilde{\sigma}^2 \to \sigma^2$

As $n \to \infty$, $\frac{n-1}{n} \to 1$, $\tilde{\sigma}^2 \to \sigma^2$

$\hat{\sigma}^2 = \frac{n-1}{n}\tilde{\sigma}^2 \xrightarrow{P} \sigma^2$ (Shown)

(7) Hypothesis Testing

Hypothesis Testing

- statistical method use to decide, based on sample data, if there is enough evidence to support or reject a claim about a population

(i) Two Sided Hypothesis Testing

$H_0: \mu = \mu_0$ (value of true population mean initially [Baseline condition])

$H_1: \mu = \mu_0$

- Assume Normality and if H_0 is true

$Y_i \overset{iid}{\sim} \text{Normal}(\mu_0, \sigma^2)$ can be represented as $\bar{Y} \sim \text{Normal}(\mu_0, \frac{\sigma^2}{n})$

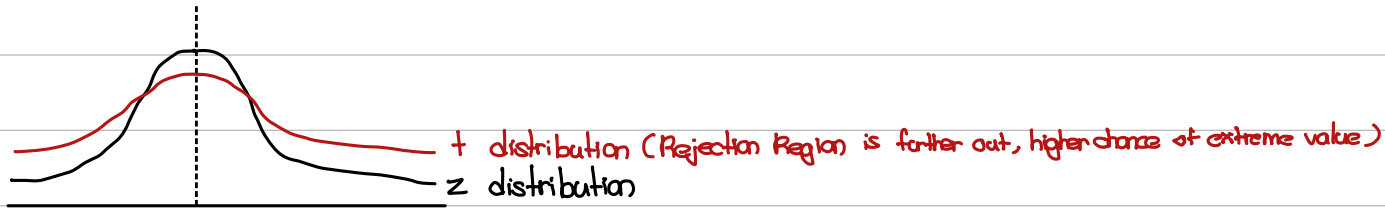
- If $\hat{\sigma}^2$ is known [Z-Test test statistic]

Test statistic: $\frac{\bar{Y} - \mu_0}{\sqrt{\sigma^2/n}} \sim N(0,1)$

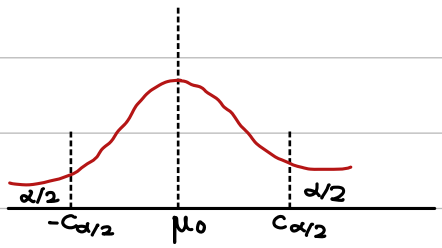
- If $\hat{\sigma}^2$ is unknown [t-Test test statistic]

Test statistic: $\frac{\bar{Y} - \mu_0}{\sqrt{\hat{\sigma}^2/n}} \sim t(n-1)$

- Graphical Representation



- Rejection Zone



whereby, $c_{\alpha/2}$ are critical values

Method 1: Critical values; test statistic rule

Reject H_0 if $|t| > t_{\alpha/2}$

Method 2: p-value rule

Reject H_0 if p-value $< \alpha$

whereby $p = 2 \times P(T > |t|)$

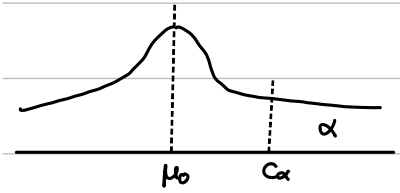
(ii) One-Sided Hypothesis Testing

$H_0: \mu \leq \mu_0$

$H_1: \mu \geq \mu_0$

- Reject H_0 if p-value $< \alpha$

Reject H_0 if $t > c_\alpha$



Asymptotic Normality

- as $n \longrightarrow \infty$, t converges to the Normal $(0,1)$

Linderberg – Levy Central Limit Theorem

- Given $\{Y_i\}_{i=1}^{n}$ are iid with $E(Y_i) = \mu$ and $Var(Y_i) = \sigma^2 < \infty$ for all i

t test statistic $= \dfrac{(\bar{Y} - \mu)}{\sqrt{\hat{\sigma}^2 / n}}$

$\sqrt{n}(\bar{Y} - \mu) \xrightarrow{d} N(0, \sigma^2)$

$\downarrow$ converge in distribution

as $n \longrightarrow \infty$, $\hat{\sigma}^2 \longrightarrow \sigma^2$

When n is large, $t \xrightarrow{a}$ Normal $(0,1)$ even when Y_i is not normally distributed

(8) Log Estimation

Given $\ln Y_i \overset{iid}{\sim} N(\mu, \sigma^2)$ for all i

Implication: $Y_i \overset{iid}{\sim}$ ln-Normal (μ, σ^2) for all i

$E(Y_i) = e^{\mu} e^{\pm\sigma^2}$

$Var(Y_i) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$

$Median(Y_i) = e^{\mu}$